



General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	03/07/2025
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	03/07/2026

Signed Off By	Oraib AlKheetanRojas
Signed Off On	19/05/2026
Sign Off Comments	

Operation Assessed	Underwater Inspection ROV – EXRAY Operation
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Associated with specific area

-  Basement 2
-  B2 029 (Robotics Lab)
-  CTP

Description of work area and/or activity assessed

Operation of the Hydromea EXRAYTM remotely-operated vehicle (ROV) for detailed underwater inspections in laboratory test tanks or controlled field sites. Activities include piloting via tether or with wireless fly-out module, recording video and sonar data, and manipulating objects with optional tools such as a gripper or vacuum cleaner. The system comprises a 9 kg neutrally-buoyant ROV, swappable lithium-ion battery packs, a 100 m low-drag tether, and topside pilot/inspector stations.

Reason for Risk Assessment

To document and control any health, safety, and equipment-damage risks introduced by actively using the EXRAY ROV in research and demonstration activities.

Overall Current Risk

Moderate Risk



Average Number of Persons Affected

2	 Employees
2	 Students

Current Rating	Hazard Information	Task Status
Moderate Risk 	Electricity The ROV operates from 14.4 V lithium-ion batteries (max 16.8 V) and can be charged in-vehicle; the tether carries power and data to topside stations. Damaged insulation, wet connectors, or incorrect charging may cause mild electric shock, short-circuit, or sparks.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Batteries limited to safety extra-low voltage (SELV) levels - Visual inspection of connectors before and after dives "Power-on last" rule for tether connection - Use of IP-rated connectors and strain relief - Operator training in safe charging and handling	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Fire Internal lithium-ion packs (97.92 Wh each, up to three) may overheat, vent, or ignite if physically damaged, over-discharged, or charged incorrectly. Fire could propagate to surrounding equipment.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Cells enclosed in sealed battery modules with integrated BMS - Charging conducted under supervision on flame-retardant surface - Fire-rated container available for damaged packs - Nearby CO2 or foam extinguisher and trained personnel	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Moving Parts Eight rim-driven hubless thrusters and any attached tools (e.g., gripper) rotate or actuate rapidly, posing impact and pinch risks to hands or loose clothing during launch, recovery, or bench tests.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Thruster guards fitted and inspected before use • Keep hands clear while motors are armed • Software and hardware emergency-stop functions available • Low-speed arm/disarm sequence in NAVIA™ interface	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Entrapment/Entanglement The 100 m tether and vehicle protrusions can snag on personnel or objects, causing trips, entanglement, or sudden pulls if tension changes.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Tether managed through reel and fair-lead • Clear deck area marked and kept free of obstacles • Gloves worn when paying out or recovering tether • Dedicated tender monitors tether tension	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Drowning / Asphyxiation Personnel working at the tank edge or launching/recovering the ROV could slip or be pulled in by the tether and lose situational awareness, leading to potential drowning, especially when wearing heavy clothing or working alone.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Minimum two-person rule during operations • Guard-rails or removable grates around deep tanks where practical • No leaning over water while the ROV is powered or tether is under tension • Throw-line or life ring kept within 5 m of launch site; staff briefed on use • For field deployments: operators wear CE-approved personal-flotation devices (PFDs) and non-slip footwear; designate a safety spotter	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

